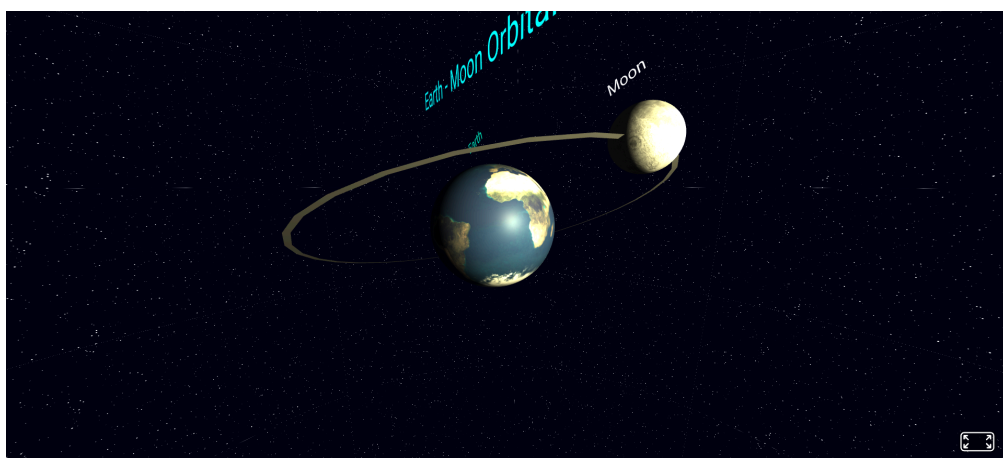


Assignment 4 – WebXR Earth-Moon Orbital System

summary:

The project was developed using HTML and the A-Frame framework to create a WebXR-based Earth-Moon orbital simulation. The scene contains realistic Earth, Moon, and Sun models created using <a-sphere> entities with high-resolution texture mapping. A space environment was designed using a sky texture, ambient lighting, and a glowing Sun that acts as the primary light source in the scene. The Earth continuously rotates on its own axis, while the Moon simultaneously rotates on its own axis and revolves around the Earth along a circular orbital path. Nested animations were used to achieve simultaneous motion. Additional features such as orbit rings, labels, camera controls, and WASD navigation were implemented to improve interactivity and visual presentation. The project successfully demonstrates WebXR concepts including 3D modeling, texture mapping, lighting, hierarchical animation, and immersive scene navigation.

Visuals



WebXR concepts used:

The project used several WebXR and A-Frame concepts to create an interactive 3D orbital simulation. A-Frame entities such as `<a-scene>`, `<a-entity>`, `<a-sphere>`, `<a-light>`, and `<a-sky>` were used to build the virtual environment and celestial bodies. Texture mapping concepts were applied by wrapping realistic Earth and Moon textures around sphere geometries using image assets.

Animation concepts were implemented using A-Frame's animation component to create continuous Earth rotation, Moon rotation, and Moon revolution. Hierarchical transformations and nested entities were used to achieve simultaneous orbital and rotational motion of the Moon around the Earth.

Lighting concepts such as ambient lighting and point lighting were used to simulate illumination from the Sun. Camera interaction concepts including look-controls and WASD movement were implemented to allow immersive navigation inside the 3D scene. The project also demonstrated real-time rendering, 3D scene management, and interactive WebXR environment creation using browser-based technologies.

Step by step technical approach:

- The project was developed using the A-Frame framework to create a 3D WebXR Earth-Moon orbital simulation. A space environment was created using `<a-scene>` and `<a-sky>` elements, while camera controls and WASD navigation enabled interaction inside the virtual environment.
- Textures for the Earth, Moon, Sun, and space background were imported using the `<a-assets>` section and applied to sphere models using the `src` attribute. The Earth was created using an `<a-sphere>` entity with a continuous self-rotation animation and a 23.5-degree axial tilt.
- The Moon was created using another `<a-sphere>` entity with its own rotation animation. Nested entities were used to make the Moon revolve around the Earth while simultaneously rotating on its own axis. Additional features such as lighting, orbit rings, labels, and a glowing Sun were added to improve realism and presentation.
- The project successfully demonstrated WebXR concepts including 3D modeling, texture mapping, lighting, animation, and hierarchical transformations using A-Frame.